Precursor Groups and the Inhour Equation

NE 630 | Nuclear Reactor Theory

28 LESSON

Read: FNRP 5.4–5.5, especially Table 5.1.

Construct a grouped-precursor model, derive the inhour equation, and compare reactor periods from one- and six-group models.

1 From one nuclide to precursor groups

Many nuclides produce delayed neutrons. For thermal fission of U-235, the ten examples below account for about 72% of delayed neutrons in the cited evaluation. The share columns give each nuclide's percentage of the total delayed-neutron yield, not a percentage of all fission neutrons.

Precursor	$t_{1/2}$ (s)	Share (%)	Precursor	$t_{1/2}$ (s)	Share (%)
^{137}I	24.6	16.3	^{85}As	2.02	5.3
^{89}Br	4.34	11.1	^{138}I	6.25	4.7
^{94}Rb	2.70	9.3	^{139}I	2.28	3.5
^{90}Br	1.91	7.4	^{95}Rb	0.378	3.5
^{88}Br	16.3	7.3	^{87}Br	55.6	3.2
Subtotal (% of delayed yield) = 71.6					

Yield shares: P. Dimitriou et al., *Nuclear Data Sheets* 173, 144–238 (2021), Table VII, U-235 thermal, CRP column (CRP+ENDF/B-VIII.0 decay data; JEFF-3.1.1 fission yields). Half-lives rounded from Tables I–II; As-85 from B. Singh and J. Chen, ENSDF (2014).

Average by delayed-neutron yield. Let s_j be the delayed-yield share of nuclide j. Normalize the shares over the precursors being averaged:

$$\bar{t}_{1/2} = \frac{\sum_j s_j t_{1/2,j}}{\sum_j s_j}, \qquad \bar{\lambda} = \frac{\ln 2}{\bar{t}_{1/2}}.$$

This preserves the mean delay $\bar{t}_{1/2}/\ln 2$. For the ten listed nuclides, divide by 71.6: $\bar{t}_{1/2} = 11.66$ s and $\bar{\lambda} = 0.05946\ \text{s}^{-1}$ *for this subset.* The remaining 28.4% is needed for a full U-235 average.

Averaging procedure: FNRP Eqs. (5.32)–(5.34).

Represent the combined release of all precursors with G effective groups, each with delayed fraction β_i and decay constant $\lambda_i = \ln 2/t_{1/2,i}$. The six-group example ($G = 6$) below is from FNRP Table 5.1.

		Delayed fraction β_i		
i	$t_{1/2,i}$ (s)	U-233	U-235	Pu-239
1	56	0.00023	0.00021	0.00007
2	23	0.00078	0.00142	0.00063
3	6.2	0.00064	0.00128	0.00044
4	2.3	0.00074	0.00257	0.00069
5	0.61	0.00014	0.00075	0.00018
6	0.23	0.00008	0.00027	0.00009
$\beta = \sum_i \beta_i$		0.00261	0.00650	0.00210
$\bar{\nu}$ (neutrons/fission)		2.50	2.43	2.90

Each C_i is an **effective delayed-neutron precursor concentration**: a reservoir of expected future delayed neutrons. Unlike Lesson 27's $C = f_n N_{\text{Br}}$, a grouped C_i alone does not determine an individual nuclide's concentration.

2 Equations and initial conditions

With $\beta = \sum_{i=1}^{G} \beta_i$,

$$\boxed{\begin{aligned} \frac{dn}{dt} &= \frac{\rho(t) - \beta}{\Lambda} n + \sum_{i=1}^{G} \lambda_i C_i + S(t), \\ \frac{dC_i}{dt} &= \frac{\beta_i}{\Lambda} n - \lambda_i C_i, \qquad i = 1, \ldots, G. \end{aligned}}$$

For an initially critical, source-free equilibrium,

$$\rho(0^-) = 0, \qquad n(0) = n_0, \qquad \boxed{C_i(0) = \frac{\beta_i n_0}{\lambda_i \Lambda}.}$$

These equations have the matrix form $\dot{\mathbf{y}} = \mathbf{A}(\rho)\mathbf{y} + \mathbf{b}$, where $\mathbf{y} = [n, C_1, \ldots, C_G]^T$ and $\mathbf{b} = [S, 0, \ldots, 0]^T$. The $(G+1) \times (G+1)$ matrix (7×7 for $G = 6$) has

$$A_{00} = (\rho - \beta)/\Lambda, \quad A_{0i} = \lambda_i,$$
$$A_{i0} = \beta_i/\Lambda, \qquad A_{ii} = -\lambda_i$$

for $i = 1, \ldots, G$, and all other entries zero.

Check. With U-235, $n_0 = 10^8$ cm^{-3}, and $\Lambda = 5 \times 10^{-5}$ s, calculate $C_1(0)$. Why is setting all $C_i(0) = 0$ a different physical experiment?

$$C_1(0) = _____ \text{ cm}^{-3}.$$

3 The pathway to the inhour equation

After a step insertion, take ρ constant and $S = 0$. The coefficients are constant; try one exponential mode:

$$n(t) = ae^{\omega t}, \qquad C_i(t) = b_i e^{\omega t}.$$

Substitute $n(t)$ and $C_i(t)$ into the precursor equations, and eliminate b_i:

$$(\omega + \lambda_i) b_i = \frac{\beta_i}{\Lambda} a \quad \Longrightarrow \quad b_i = \frac{\beta_i a}{\Lambda(\omega + \lambda_i)}.$$

Substitution into the neutron equation gives

$$\Lambda\omega = \rho - \beta + \sum_{i=1}^{G} \frac{\beta_i \lambda_i}{\omega + \lambda_i}.$$

Use $\beta = \sum_i \beta_i$ and rearrange to obtain

$$\boxed{\rho = \Lambda\omega + \sum_{i=1}^{G} \frac{\beta_i \omega}{\omega + \lambda_i}.}$$

This is the **inhour equation**. It is rational; clearing the denominators gives the degree-$(G+1)$ characteristic polynomial. For positive β_i and distinct λ_i, as here, the poles $-\lambda_i$ are not roots.

The full solution is a sum of $G + 1$ modes (seven for six groups), with amplitudes fixed by the initial state. The largest eigenvalue $\omega_\star$ determines the late-time signed *stable period*, $T = 1/\omega_\star$:

$$\boxed{\rho = \frac{\Lambda}{T} + \sum_{i=1}^{G} \frac{\beta_i}{1 + \lambda_i T}.}$$

At criticality, $\omega_\star = 0$ and the equilibrium period is infinite. For a negative insertion, $T < 0$.

Reactor periods and a numerical comparison

4 One effective group and useful limits

Near criticality, if $|\omega_\star| \ll \lambda_i$ for every group,

$$T \simeq \frac{\Lambda + \sum_i \beta_i/\lambda_i}{\rho}.$$

A one-group model using Section 1's average preserves the equilibrium total precursor inventory:

$$\beta_{1g} = \beta, \qquad \lambda_{1g} = \bar{\lambda} = \frac{\beta}{\sum_i \beta_i/\lambda_i}.$$

For the U-235 groups on page 1, $\bar{t}_{1/2} = 9.044$ s and $\bar{\lambda} = 0.076641\ \text{s}^{-1}$. This reproduces the small-reactivity limit. A mixture still has several modes; one effective group cannot reproduce every transient. This is distinct from L27's Br-87-only model.

Well above prompt criticality,

$$T \simeq \frac{\Lambda}{\rho - \beta}.$$

This estimate requires the delayed term to be small relative to $\rho - \beta$; the full period is finite at $\rho = \beta$.

For a sufficiently large negative insertion,

$$\omega_\star \longrightarrow -\lambda_1, \qquad T \longrightarrow -\frac{1}{\lambda_1} = -80.79\ \text{s}.$$

Earlier decay can be much faster than this late-time limit.

5 Work with `ne630_kinetics`

Save `ne630_kinetics.py` beside your notebook or script. Its two functions are below. `f` is the right side of Section 2's equations for $\mathbf{y} = [n, C_1, \ldots, C_G]^T$, with $S = 0$ and no feedback. `solve` starts from critical equilibrium and passes `f` to SciPy's `solve_ivp`.

```
import numpy as np
from scipy.integrate import solve_ivp

def f(t, y, rho, betas, lambdas, Lambda):
    n = y[0]            # Unpack the unknowns so that
    C = y[1:]           #   the equations are clearer.
    beta = sum(betas)  # Total delayed neutron fraction
    dndt = ((rho(t)-beta)/Lambda)*n + sum(C*lambdas)
    dCdt = -C*lambdas + betas*n/Lambda
    dydt = np.zeros(len(y))
    dydt[0] = dndt      # Collect the derivatives
    dydt[1:] = dCdt     #   into one array that we return
    return dydt


def solve(rho, times, n0, betas, lambdas, Lambda):
    """Return n(t) and C_i(t), starting critical at n0.

    rho is a function of t, and C[i] is group i+1.
    """
    y0 = np.zeros(len(betas) + 1)
    y0[0] = n0
    y0[1:] = betas * n0 / (lambdas * Lambda)
    # Limit the step to the spacing of times so that the
    # solver cannot step right over a change in rho.
    hmax = np.max(np.diff(times))
    sol = solve_ivp(f, (times[0], times[-1]), y0,
                    args=(rho, betas, lambdas, Lambda),
                    t_eval=times, max_step=hmax,
                    method="LSODA", rtol=1e-8)
    return sol.y[0], sol.y[1:]
```

`solve` returns `n` at each time in `times` and `C`, where `C[i]` is C_{i+1}; with $n_0 = 1$, these are n/n_0 and C_i/n_0. The argument `rho` is a Python function of t (s) that returns the dimensionless reactivity, so d dollars is $\rho = d\beta$. Enter the U-235 data from page 1. For one group, pass $[\beta]$ and Section 4's $[\bar{\lambda}]$; `solve` builds that model's own critical precursor inventory.

```
import numpy as np
from ne630_kinetics import solve

betas = np.array([0.00021, 0.00142, 0.00128,
                  0.00257, 0.00075, 0.00027])
thalfs = np.array([56, 23, 6.2, 2.3, 0.61, 0.23])  # s
lambdas = np.log(2)/thalfs
Lambda = 5e-5                          # s
beta = sum(betas)
d = 0.2                                # dollars

def rho(t):
    return d*beta                      # step at t = 0

times = np.linspace(0, 200, 2001)    # s
n6, C6 = solve(rho, times, 1.0, betas, lambdas, Lambda)
b1, l1 = np.array([beta]), np.array([0.076641])
n1, C1 = solve(rho, times, 1.0, b1, l1, Lambda)
```

Once the faster modes have decayed, $\ln n$ is a straight line with slope $1/T$, so $T \simeq (t_2 - t_1)/\ln[n(t_2)/n(t_1)]$ for two late times. Because `hmax` is the spacing of `times`, choose `times` finely enough to catch every change in ρ. From a terminal, `python ne630_kinetics.py` plots Lesson 27's Example 4.

6 Compare one group with six

Verify equilibrium. Set $\rho = 0$: all densities should remain constant. Inspect `f` against Section 2 and check that evaluating it at the critical equilibrium gives zero derivatives.

Compute periods. Run $\rho/\beta = 0.02$ and 0.2. Compare the six-group period, the effective one-group period, and the small-reactivity estimate in Section 4.

ρ/β	T_{6g} (s)	T_{1g} (s)
0.02		
0.2		

Substitute each six-group frequency $\omega = 1/T$ into the inhour equation in Section 3 and check that its right-hand side equals the imposed ρ.

Plot and explain. Plot both 0.2β traces over 0–200 s on semilog axes. Compare the late slope with $1/T$. Why does preserving the mean delay and equilibrium inventory fail to preserve the whole transient? Which insertion is better represented by one effective group?

Solution verification. Whenever we solve an engineering problem, we are responsible for verifying the correctness of our solution. How can we verify that our computed neutron densities and precursor concentrations are correct?